Hints: Functional sequences, series and spaces – 1. Sequences of functions and uniform convergence

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. The supremum criterion is just the definition unpacked: "for every ε there is N such that $|f_n(x) - f(x)| \leqslant \varepsilon$ for all x and $n \geqslant N$" is the same as $\sup_x |f_n - f| \leqslant \varepsilon$ for $n \geqslant N$. For the Cauchy criterion, the nontrivial direction: the Cauchy condition makes each $f_n(x)$ a Cauchy sequence, so define f pointwise, then let $m \to \infty$ in $|f_n(x) - f_m(x)| \leqslant \varepsilon$ with x fixed.

Hint 2. Continuity: the $\varepsilon/3$ argument $|f(x) - f(y)| \leqslant |f - f_n|(x) + |f_n(x) - f_n(y)| + |f_n - f|(y)$. Integrability: if $\sup|f - f_n| \leqslant \varepsilon$, then for every partition $U(f,P) - L(f,P) \leqslant U(f_n,P) - L(f_n,P) + 2\varepsilon(b-a)$, and $|\int f_n - \int f| \leqslant \varepsilon(b-a)$.

Hint 3. Derivatives: write $f_n(x) = f_n(x_0) + \int_{x_0}^x f_n'$ and pass to the limit (the integral part converges uniformly on a bounded interval by the previous part); then $f(x) = f(x_0) + \int_{x_0}^x g$ with g continuous, so $f' = g$. For the zoo: x^n and nxe^{-nx^2} do not converge uniformly (find the moving point: $x = 1 - 1/n$, resp. $x = 1/\sqrt{2n}$ where the value is $\sqrt{n/2}\, e^{-1/2}$); $\frac{\sin nx}{\sqrt{n}}$ converges uniformly but $f_n' = \sqrt{n}\cos nx$ does not converge; for $\frac{x}{1+n^2x^2}$ the derivatives converge only pointwise (limit 1 at 0, 0 elsewhere), and indeed $f' = 0 \neq g(0)$.

Problem 2

Hint 1. Necessity: a precompact set is totally bounded; take a finite ε-net $g_1, \ldots, g_k$ in $\mathcal{F}$. Boundedness is immediate, and equicontinuity follows from the (uniform) continuity of the finitely many g_i plus the $\varepsilon/3$ trick.

Hint 2. Sufficiency: given a sequence in $\mathcal{F}$, take a countable dense set $\{x_j\} \subset K$ and use a diagonal argument (each $(f_n(x_j))_n$ is bounded) to get a subsequence converging at every x_j.

Hint 3. Show that this subsequence is uniformly Cauchy: on a compact set equicontinuity is uniform (one δ for all points and all f), cover K by finitely many δ-balls around points x_j of the dense set, and compare $f_n(x)$ with $f_n(x_j)$. Counterexamples: x^n or $\sin nx$ (bounded, not equicontinuous, no uniformly convergent subsequence), and the constants $f_n \equiv n$ (equicontinuous, unbounded).

Problem 3

Hint 1. For uniform convergence the answer is yes: an $\varepsilon/3$ argument, choosing first n with $\sup|f_n - f| < \varepsilon/3$ and then δ from the uniform continuity of that single f_n.

Hint 2. For pointwise convergence the answer is no. Truncate a function that is continuous but not uniformly continuous on $\mathbb{R}$, e.g. $f_n(x) = \min(x^2, n)$: each f_n is Lipschitz, and $f_n \to x^2$ pointwise.

Problem 4

Hint 1. First pass to a subsequence in which all functions are monotone in the same direction (infinitely many are nondecreasing or infinitely many are nonincreasing). Then use a diagonal argument to get a subsequence converging at every rational point of $[a,b]$ (and at a, b).

Hint 2. The limit φ on the rationals is monotone; extend it to a monotone function g on $[a, b]$. At every continuity point x of g, squeeze $f_n(x)$ between $f_n(r)$ and $f_n(s)$ with rationals $r < x < s$ close to x to show $f_n(x) \to g(x)$.

Hint 3. A monotone g has at most countably many discontinuities; one more diagonal argument handles them. The limit is monotone: a pointwise limit of nondecreasing functions is nondecreasing (after the first reduction all f_n have the same direction).

Problem 5

Hint 1. Compute $f_n(1)$ first: the product telescopes to $n + 1$, so $f_n(1) = 1$. Also note each f_n is increasing in x and nonnegative on $[0, 1]$.

Hint 2. Use $1 + t \leqslant e^t$: $\prod_{k=1}^{n}(1 + x/k) \leqslant e^{xH_n} \leqslant e^x\, n^x$, where $H_n \leqslant 1 + \ln n$. Hence on $[0, a]$ with $a < 1$ we have $0 \leqslant f_n \leqslant f_n(a) \leqslant e\, n^a/(n + 1) \to 0$, which is uniform convergence. The pointwise limit is 0 on $[0, 1)$ and 1 at $x = 1$ (so the convergence is not uniform on $[0, 1]$).